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EXAMINER
HILGENDORF, DALE W

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3662	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/843,546	Applicant(s) Wu et al.	
	Examiner DALE W HILGENDORF	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 9/26/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-6,8-15 and 17-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-6,8-15 and 17-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 6/17/2022 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Amended claims 1 thru 6, 8 thru 15 and 17 thru 20 have been entered into the record. Claims 7 and 16 have been cancelled.

Response to Amendment

3. The amendments to the specification overcome all but one of the drawing objections from the previous office action (6/26/2024). All of the drawing objections are withdrawn with the exception of the drawing objection recited below in this office action. See the response to arguments for further explanation of the objection.

4. The amendments to the specification overcome the specification objections from the previous office action (6/26/2024). The specification objections are withdrawn.

5. The amendments to the claims overcome the prior art rejections from the previous office action (6/26/2024). The prior art rejections are withdrawn.

Response to Arguments

6. Applicant's arguments filed 9/26/2024 regarding the drawing objection to the P and Q notations, have been fully considered but they are not persuasive. The applicant states that a person of ordinary skill in the art would not be confused by the description of Figure 5 in reference to P[0051] thru P[0053], and that there is not suggestion that P or Q might denote a probability of success or a probability of failure (argument page 13). The examiner respectfully disagrees based on the content of the disclosure. The disclosure is directed to probability distribution patterns (applied to autonomous driving).

In the field of mathematical probabilities, the notations of “P” and “Q” are the common and well known notations for “Probability of Success” and Probability of Failure”, respectively. A simple Google search of “distribution pattern notations probability of success P failure Q” returned the results of:

> In probability notation, "P" represents the probability of success, while "Q" represents the probability of failure, with the relationship being: Q = 1 - P; meaning that the probability of failure is always calculated by subtracting the probability of success from 1.
>
> Key points about P and Q:
>
> P (Probability of Success): This is the likelihood of a desired outcome occurring in a given trial or experiment.
>
> Q (Probability of Failure): This is the likelihood of the undesired outcome happening in a trial.

The confusing part about the references to P and Q are whether they are merely variables selected by the applicant, or they are directed to the probability notations that are commonly used in the art.

7. Applicant's arguments filed 9/26/2024 regarding the interpretation of the "processing device" of claim 1, have been fully considered but they are not persuasive. The interpretation of the processing device is directed to claim 1, not to claim 19. Claim 19 is used as the basis of interpretation of the processing device of claim 1 because the specification does not provide direct reference to the processing device being a processor. Claim 1 recites, “generating, by a processing device, a controlled trajectory” and “computing, by the processing device, a similarity score”. These limitations invoke

35 U.S.C. 112(f). The claimed "device" is a generic placeholder substitute for "means" (Prong A of 112f analysis MPEP 2181). The claim limitations are directed to generating a controlled trajectory and computing a similarity score by processing (Prong B of 112f analysis). And, claim 1 is not modified by any structure to achieve the specified function (Prong C of 112f analysis).

8. In order "to determine whether a word, term, or phrase coupled with a function denotes structure, examiners may check whether: (1) the specification provides a description sufficient to inform one of ordinary skill in the art that the term denotes structure; (2) general and subject matter specific dictionaries provide evidence that the term has achieved recognition as a noun denoting structure; and/or (3) the prior art provides evidence that the term is an art-recognized structure to perform the claimed function. Ex parte Rodriguez, 92 USPQ2d 1395, 1404 (Bd. Pat. App. & Int. 2009) (precedential)" (MPEP 2181.I.C). The applicant's specification did not provide a description of the claimed "processing device" so the examiner used the claimed language of claim 19 where the processing device was defined to include a processor and a memory. The general and subject matter specific dictionaries provide a definition of a "processing device" of the hardware that is inside a computer to include the processor and memory. The applicant argues that the term "processing device" is known in the art and provides the structure because of it be well known. The examiner equates the processing device being known in the art to prior art that provides evidence that the term is an art-recognized structure to perform the claimed function. Each of these ways to determine whether a word, term, or phrase coupled with a function denotes structure requires that 112f is invoked. Therefore, the interpretation under 35

U.S. C. 112(f) remains invoked as recited in the previous office action (6/26/2024) and as restated below in this office action.

9. Applicant's arguments filed 9/26/2024 regarding the 35 U.S.C. 101 rejections have been fully considered but they are not persuasive. The applicant argues that the amended independent claims now recite an improvement to the technology of autonomous driving (argument pages 15 and 16). The examiner respectfully disagrees because the claimed are not specifically directed to autonomous driving. The claims recite that they generate a trajectory of an autonomous driving vehicle (ADV) in a scenario and the trajectory is executable to drive the scenario. There is no positive recitation of actual control of the vehicle, merely generating a trajectory that is executable. This appears to be generating program instructions, which are merely used to provide information to the ADV, not actual control of the ADV.

10. The independent claims further recite that the method receives data and identifies a pattern from the data, then performs computations to determine first and second trajectories. Then based on the computations, the method discards (removes or filters out) one of the trajectories. The computing limitations are simply the performing of mathematical operations or formulas (also an abstract idea). A person can perform such computations, either mentally, with pen and paper, or with the use of a general purpose computer. Because there is no actual control of the vehicle, there is not a need for the computations to be performed in a “real-time” situation. Simply put, claim 1 computes two trajectories and discards one of them. These amended portions of the independent claims are also part of the abstract idea and do not recite a practical application of the abstract idea.

11. The applicant further argues that the amended claims are eligible based on the 101 example 47 claim 3. The decision to not use computed data is not the same as the blocking of signal reception (example 47, claim 3). The claimed identifying, computing and discarding may all be performed mentally, even though the claim language implies that they are performed by the processing device. The claimed processing device is merely using a computer as a tool to perform the abstract idea. In discussion of the claim limitations with a subject matter eligibility expert, the expert agreed with the examiner to maintain the 101 rejections of the claims. The performing of various computations and then discarding some of the results of the computations is not an improvement to the functioning of a computer, or an improvement to any other technology or technical field. The subject matter expert (and the examiner) suggests a claim limitation directed to controlling the ADV based on the first controlled trajectory (non-discarded trajectory) in order to overcome the 101 rejections.

Drawings

12. The drawings are objected to because in Figure 5, the probability distribution 520 is associated with function p(x) and the probability distribution 530 is associated with function q(x). But in the specification P[0052] and P[0053, the "probability distribution 530 (P)" and the "probability distribution 520 (Q)" are recited. It is unclear if the P and Q are meant to be equated to the p(x) and q(x) of Figure 5, or if the P denotes a probability of success for 530 and Q denotes a probability of failure of 520. If they reference Figure 5, then the P and Q should refer to the other probability distribution. If they are the probability of success and failure, such explanation should be included in the specification (to avoid confusion with the functions p and q of Figure 5). In any

case, a further explanation of P and Q are required in reference to the specification formulas or Figure 5 or the general definition of P and Q. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Interpretation

13. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and

such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

14. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim

limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

15. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are: generating and computing by a processing device in claim 1.

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof. The claimed processing device is interpreted as a processor and memory based on the recited structure of claim 19.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 101

16. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

17. Claims 1 thru 6, 8 thru 15 and 17 thru 20 are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea without significantly more.

Subject Matter Eligibility Criteria - Step 1:

18. Claim 1 is directed to a method (i.e., a process). Accordingly, claim 1 is within at least one of the four statutory categories. Claim 10 is directed to a non-transitory computer-readable storage medium (i.e., a machine). Accordingly, claim 10 is within at

least one of the four statutory categories. Claim 19 is directed to a processing device (i.e., a machine). Accordingly, claim 19 is within at least one of the four statutory categories.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2A - Prong One:

19. Regarding Prong One of Step 2A of the Alice/Mayo test (which collectively includes the guidance in the January 7, 2019 Federal Register notice and the October 2019 update issued by the USPTO as now incorporated into the MPEP, as supported by relevant case law), the claim limitations are to be analyzed to determine whether, under their broadest reasonable interpretation, they "recite" a judicial exception or in other words whether a judicial exception is "set forth" or "described" in the claims. MPEP 2106.04(II)(A)(1). An "abstract idea" judicial exception is subject matter that falls within at least one of the following groupings: a) certain methods of organizing human activity, b) mental processes, and/or c) mathematical concepts. MPEP 2106.04(a).

20. Independent claim 1 includes limitations that recite at least one abstract idea. Specifically, independent claim 1 recites:

A computer-implemented method comprising:

generating, by a processing device, a controlled trajectory of an autonomous driving vehicle (ADV) in a scenario, the controlled trajectory executable by the ADV to drive autonomously in the scenario;

receiving a set of data acquired in a plurality of driving demonstrations in the scenario;

identifying a distribution pattern of the set of data acquired, the distribution pattern indicating probabilities of driving trajectories in the scenario;

computing, by the processing device, a similarity score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations;

computing a first mean square error and a first similarity score between a first controlled trajectory generated by the processing device and the set of data of the plurality of driving demonstrations in the scenario;

computing a second mean square error and a second similarity score between a second controlled trajectory generated by the processing device and the set of data of the plurality of driving demonstrations in the scenario, wherein the first mean square error is greater than the second mean square error while the first similarity score is higher than the second similarity score; and

discarding the second controlled trajectory and parameters associated therewith.

21. The above underlined limitation constitutes “a mental process” because it is an observation/evaluation/judgment/analysis that can, at the currently claimed high level of generality, be practically performed in the human mind (e.g., with pen and paper). For instance, a person could decide on a path for the vehicle to follow. This typically occurs when a person is driving a car, such as drive straight to the intersection, stop at the stop sign, and turn right when clear and you have the right of way. A person can use this knowledge even when they are not driving, just planning a trip. A person can then use acquired data (different options for the scenario) and look at the scenario data to identify patterns. A person can then determine which of the acquired scenarios are closest to

their plan for the trip, make computations of different controlled trajectories and then get rid of one of the computed trajectories. See P[0022] of the present application which also disclose that the evaluations of the planning and control are based on human driving. The examiner notes that the claimed "controlled trajectory executable by the ADV to drive autonomously" is merely that it is able to be used and does not positively recite any control of the ADV. Accordingly, the claim recites at least one abstract idea.

22. Claims 10 and 19 recite the same limitations as the method steps of claim 1, with the addition of non-transitory machine-readable medium having instructions stored and executed by a processor to cause the processor to perform operations (claim 10), and a processor and a memory coupled to the processor storing instructions executed by the processor (claim 19). These claims also recite at least one abstract idea.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2A - Prong Two:

23. Regarding Prong Two of Step 2A of the Alice/Mayo test, it must be determined whether the claim as a whole integrates the abstract idea into a practical application. As noted at MPEP §2106.04(II)(A)(2), it must be determined whether any additional elements in the claim beyond the abstract idea integrate the exception into a practical application in a manner that imposes a meaningful limit on the judicial exception. The courts have indicated that additional elements such as merely using a computer to implement an abstract idea, adding insignificant extra solution activity, or generally linking use of a judicial exception to a particular technological environment or field of use do not integrate a judicial exception into a "practical application." MPEP §2106.05(I)(A).

24. In the present case, the additional limitations beyond the above-noted at least one abstract idea recited in claim 1 are as follows (where the bolded portions are the "additional limitations" while the underlined portions continue to represent the at least one "abstract idea"):

A computer-implemented method comprising:

generating, by a **processing device** (using computers or machinery as mere tools to perform the abstract idea as noted below, see MPEP § 2106.05(f)), a controlled trajectory of an **autonomous driving vehicle (ADV)** (the machine is merely an object on which the method operates to perform the abstract idea as noted below, see MPEP § 2106.05(b)) in a scenario, the controlled trajectory executable by the ADV to drive autonomously in the scenario;

receiving a set of data acquired in a plurality of driving demonstrations in the scenario (extra-solution activity (data gathering) as noted below, see MPEP § 2106.05(g));

identifying a distribution pattern of the set of data acquired, the distribution pattern indicating probabilities of driving trajectories in the scenario;

computing, by the **processing device** (using computers or machinery as mere tools to perform the abstract idea as noted below, see MPEP § 2106.05(f)), a similarity score based on comparisons between the controlled trajectory and the distribution pattern of the plurality of driving demonstrations;

computing a first mean square error and a first similarity score between a first controlled trajectory generated by the processing device and the set of data of the plurality of driving demonstrations in the scenario;

computing a second mean square error and a second similarity score between a second controlled trajectory generated by the processing device and the set of data of the plurality of driving demonstrations in the scenario, wherein the first mean square error is greater than the second mean square error while the first similarity score is higher than the second similarity score; and

discarding the second controlled trajectory and parameters associated therewith.

25. For the following reasons, the above-identified additional limitations, when considered as a whole with the limitations reciting the at least one abstract idea, do not integrate the above-noted at least one abstract idea into a practical application.

Regarding the additional limitation of a processing device, this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an

ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 1 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 1 is directed to at least one abstract idea.

26. Similarly, claim 10 recites, non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations. The operations performed are the same as the method steps of claim 1, wherein the same analysis of the abstract idea also applies.

Regarding the additional limitations of non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations, these limitations amount to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 10 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 10 is directed to at least one abstract idea.

27. Similarly, claim 19 recites, a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor, cause the processor to perform the operations that are the same as the method steps of claim 1, wherein the same analysis of the abstract idea also applies.

Regarding the additional limitation of a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor; this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of an autonomous driving vehicle (ADV), this limitation amounts to merely providing the object on which the method operates to perform the above-noted at least one abstract idea (see MPEP § 2106.05(b)).

Regarding the additional limitation of receive a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation merely adds insignificant extra-solution activity (data gathering) to the at least one abstract idea in a

manner that does not meaningfully limit the at least one abstract idea (see MPEP § 2106.05(g)).

Thus, taken alone, the additional elements do not integrate the at least one abstract idea into a practical application. Looking at the additional limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. MPEP §2106.05(I)(A) and §2106.04(II)(A)(2).

For these reasons, claim 19 does not recite additional elements that integrate the judicial exception into a practical application. Accordingly, claim 19 is directed to at least one abstract idea.

Subject Matter Eligibility Criteria - Alice/Mayo Test: Step 2B:

28. Regarding Step 2B of the Alice/Mayo test, claims 1, 10, and 19 do not include additional elements (considered both individually and as an ordered combination) that are sufficient to amount to significantly more than the judicial exception for reasons the same as those discussed above with respect to determining that the claim does not integrate the abstract idea into a practical application.

Regarding claim 1, the additional limitation of a processing device, this limitation amounts to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding claim 10, the additional limitations of non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations, these limitations amount to merely using a

computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding claim 19, the additional limitations of a processing device, comprising: a processor; and a memory coupled to the processor and storing instructions, which when executed by the processor, cause the processor to perform the operations; these limitations amount to merely using a computer or other machinery as tools performing their typical functionality in conjunction with performing the above-noted at least one abstract idea (see MPEP § 2106.05(f)).

Regarding the additional limitation of receiving a set of data acquired in a plurality of driving demonstrations in the scenario, this additional limitation has been reevaluated, and it has been determined that such limitations are not unconventional as they merely consist of data gathering which is recited at a high level of generality. See OIP Techs., Inc., v. Amazon.com, Inc., 788 F.3d 1359, 1363, 115 USPQ2d 1090, 1093 (Fed. Cir. 2015) (sending messages over a network); or buySAFE, Inc. v. Google, Inc., 765 F.3d 1350, 1355, 112 USPQ2d 1093, 1096 (Fed. Cir. 2014) (computer receives and sends information over a network). Further, adding a preliminary step of gathering data to a process that only recites identifying patterns and determining comparisons of the patterns (a mental process) does not add a meaningful limitation to the process of determining a similarity score of the scenario. See MPEP 2106.05(d)(II) and 2106.05(g).

29. The dependent claims 2 thru 6, 8, 9, 11 thru 15, 17, 18 and 20 do not provide additional elements or a practical application to become eligible under 35 U.S.C. 101. The dependent claims are directed to adjusting a parameter used for the generation of

the trajectory and updating the trajectory based on previous computations (claims 2, 11, 20), generating candidate trajectories (claims 3, 12), calculation methods for the similarity score (claims 4, 13) (also abstract idea of mathematical formulas), defining the scenario (claims 5, 6, 14, 15), defining the distribution pattern (claims 8, 17), and defining the source of the driving demonstrations (claims 9, 18). These limitations are extra-solution activity, or part of the abstract idea (mental activity and/or mathematical equations). They do not constitute a practical application of the abstract idea.

Related Art

30. The examiner points to Li et al PGPub 2018/0374359 A1 as related art, but not relied upon for any rejection. Li et al is directed to training a deep neural network and evaluating a prediction method of autonomous driving (Figures 7 and 8). More specifically, Figure 8 is directed to "operation 801, processing logic receives a predicted trajectory (e.g., unknown predicted trajectory), which was generated using a prediction method or algorithm based on perception data perceiving a driving environment surrounding an ADV. For at least some of the trajectory points of the predicted trajectory, in operation 802, processing logic extracts a set of features from the selected trajectory points. In operation 803, processing logic applies a predetermined DNN model to the features to categorize the features and generate a similarity score. The similarity score represents the similarity between the predicted trajectory and one or more prior actual trajectories. In operation 804, processing logic determines a performance or accuracy of the prediction method or algorithm based on the similarity score." P0065].

Conclusion

31. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

32. Any inquiry concerning this communication or earlier communications from the examiner should be directed to DALE W HILGENDORF whose telephone number is (571)272-9635. The examiner can normally be reached Monday - Friday 9-5:30.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jelani Smith can be reached on 571-270-3969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/DALE W HILGENDORF/
Primary Examiner, Art Unit 3662